25901278 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCFSC200 - BIOMETRICS AND IMPRESSION ANALYSIS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. Which type of fingerprint is formed by blood, ink, or paint? [U] / [CO1]
2. Which ion in sweat reacts with silver nitrate? [U] / [CO1]
3. Explain the advantage of fluorescent powders over conventional powders. [U] / [CO1]
4. Explain significance of toe alignment in identification. [U] / [CO4]
5. How would you record fingerprints from an unconscious person in a hospital? [A] / [CO3]
6. What is ridge detail? [K] / [CO3]
7. Explain the appropriate technique to lift an impression from a smooth surface. [An] / [CO4]
8. State the importance of palm prints in forensic investigation. [K] / [CO4]
9. Explain comparison procedures of test standards with crime scene impressions. [U] / [CO5]
10. What is a patent fingerprint? [U] / [CO2]

11. Ninhydrin reacts with which component of sweat? [U] / [CO1]

12. Explain differences between human and animal footprints [U] / [CO4]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

13. Explain the challenges faced while examining fingerprints from dead bodies. [U] / [CO3]

14. Describe the methods used for lifting developed fingerprints. [U] / [CO1]

15. Classify physical and chemical methods used in developing lip prints. [A] / [CO4]

16. Apply lifting methods in dry surface impressions. [A] / [CO4]

17. Write short notes on filters used in forensic photography. [K] / [CO3]

18. Explain forensic development of foot impressions [A] / [CO4]

19. Explain the forensic importance of different types of fingerprints. [U] / [CO1]

[4x5 = 20]

Part C

Essay questions

Answer any **1** questions

Each question carries **10** marks

20. Analyze forensic reconstruction using footprint sequences. [An] / [CO4]

21. Describe in detail the use of alternate light sources, oblique lighting, and filters in detecting and photographing latent fingerprints. [S] / [CO3]

[10x1 = 10]